Branch Coverage Report

Online Bookstore – SE Mini-Project (UE24CS341A), PES University, Group 2

Part 3 – Testing (2–6 November 2026)

Item	Detail
Project	Online Bookstore (multi-seller book marketplace)
Prepared by	Group 2
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Repository	SE-OnlineBookstore/SE-Mini-Project
Version	1.0

1. Purpose and scope

This report covers branch (decision) coverage of the server code from the same final run as the Code Coverage Report (33 suites, 256 tests, all passing). A branch is each outcome of an if, ternary, logical operator or default value.

2. Summary

Area	% Branch
All files	74.56
middleware	85.45
services	73.71
utils	78.94

3. Branch coverage by file (lowest first)

File	% Branch	% Stmts	Uncovered lines
requestId.js	50	100	6
orderService.js	61.15	68.22	19-25,31,40,47,57,61,65,94,99,103,107,112-113,161-284,361,372,427,477
rateLimit.js	66.66	100	23
cartService.js	66.66	79.71	77,85,94,98,102-103,120,125,129,146,154-156
crypto.js	66.66	100	4
wishlistService.js	68.18	86.36	67,74,96,114,133,136,176,186
invoiceService.js	69.23	95.42	229,258,263-264,295-296
paymentService.js	69.69	88.61	93,98,133,179,194-197,227,237,242,261

AppError.js	70	93.75	21
checkoutService.js	70.4	88.97	50,73,87,98-99,112,175,214,222
moderationService.js	75	91.3	28,35,42,49
catalogueService.js	75.94	90.82	72,143,164-165
listingService.js	79.06	89.09	15,38,67,87,113,158
errorHandler.js	80	81.81	11,15,17,24
notificationService.js	80	92.85	140-145
authService.js	82.14	93.33	63-64
fulfilmentService.js	84.44	96.29	35,75
reviewService.js	85.71	90.9	45-46
tokenService.js	85.71	96.96	34,56
validate.js	90	100	11
otpService.js	90.9	100	15,30
userService.js	95.45	100	70
authorizeRoles.js	100	100	
verifyJWT.js	100	100	
analyticsService.js	100	100	
auditService.js	100	100	
sellerDashboardService.js	100	100	
asyncHandler.js	100	100	
isbn.js	100	100	
money.js	100	0	2-6

4. Files at or above 85% branch coverage

reviewService.js, tokenService.js, validate.js, otpService.js, userService.js, authorizeRoles.js, verifyJWT.js, analyticsService.js, auditService.js, sellerDashboardService.js, asyncHandler.js, isbn.js.

5. Weakest files and plan

File	% Branch	Uncovered lines	Status
requestId.js	50	6	Known gap; not required by the NFR-06 gate.
orderService.js	61.15	19-25,31,40,47,57,61,65,94,99,103,107,112-113,161-284,361,372,427,477	Known gap; not required by the NFR-06 gate.
rateLimit.js	66.66	23	Known gap; not required by the NFR-06 gate.
cartService.js	66.66	77,85,94,98,102-103,120,125,129,146,154-156	Known gap; not required by the NFR-06 gate.

crypto.js	66.66	4	Known gap; not required by the NFR-06 gate.
wishlistService.js	68.18	67,74,96,114,133,136,176,186	Known gap; not required by the NFR-06 gate.
invoiceService.js	69.23	229,258,263-264,295-296	Known gap; not required by the NFR-06 gate.
paymentService.js	69.69	93,98,133,179,194-197,227,237,242,261	Known gap; not required by the NFR-06 gate.

6. Change during Part 3

Measure	Run of 5 Nov	Final run of 6 Nov
All files branches	72.38	74.56
Services branches	71.44	73.71
auditService.js	0	100
userService.js	36.36	95.45

7. Conclusion

Branch coverage is 74.56% overall and 73.71% for services. The project gate (NFR-06) is defined on statements only, so it is met. The two weakest service files from the earlier run (auditService and userService) were raised to 100% and 95.45%. The remaining low-branch files are listed with causes above.